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Course	Digital Signal Processing Lab	**Experiment No.**	05
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Autocorrelation

Measuring the solar cycle from 277 years of sunspot observations

Abstract: The autocorrelation of the international monthly sunspot number (3,333 months, 1749 to 2026) was computed by the direct definition, by convolution with the folded sequence and through the power spectrum (Wiener-Khinchin); the three agreed to 4e-16. The normalised autocorrelation has its first peak at 127.8 months, i.e. a solar cycle of 10.65 years (ρ = 0.579), with repeats at 2, 3 and 4 cycles whose decreasing height (0.35, 0.25, 0.19) measures the cycle-to-cycle irregularity. The series decorrelates after 36 months, the period was stable at 10.7 to 11.0 years since 1818 but only 9.3 years in the irregular 18th century, and the cycle was still detected when buried in white noise at an SNR of -10 dB. The leftover month-to-month variation is not white noise, which the 95 % confidence band reveals.

Keywords: *autocorrelation, periodicity, Wiener-Khinchin, white-noise test, sunspot number, solar cycle, space weather, Python*

1. Aim and Objectives

Aim: to study the autocorrelation of a discrete-time signal, verify its properties, and use it to detect hidden periodicity and to test for randomness in a real-world signal, in Python.

- Compute autocorrelation three ways and verify symmetry, the maximum at zero lag and r[0] = energy.
- Estimate the period of the solar cycle and how it changed over 277 years.
- Test whether the residual variation is white noise, and how much noise periodicity detection tolerates.

2. Real-World Application

The Sun's activity rises and falls in a cycle of roughly 11 years, and sunspots are its oldest and most reliable measure. Solar maximum brings more flares and geomagnetic storms, which heat and expand the upper atmosphere, increase drag on low-orbit satellites, disturb GPS and HF radio, and induce currents in power grids. Satellite operators, ISRO's Aditya-L1 mission team, airlines on polar routes and grid operators therefore plan around the cycle, and the first step is to measure its period from data. Autocorrelation does exactly that: it compares a signal with delayed copies of itself, so any repetition shows up as a peak at the repetition lag even when it is hard to see in the raw data (Table 1).

Table 1. Uses of autocorrelation

Field	Signal	What the autocorrelation reveals
Space weather	monthly sunspot number	solar-cycle period, decorrelation time, predictability
Speech processing	voiced speech frame	pitch period (fundamental frequency) of the speaker
Radar and sonar	echo of a coded pulse	sharp peak at zero lag gives fine range resolution
Machine monitoring	bearing vibration	repetition rate of a defect hitting the race
Statistics, finance	residuals of a model	whether errors are white (model adequate) or correlated

3. Theory

The **autocorrelation** of an energy signal measures its similarity with a copy shifted by l samples:

$$r_{xx}[l] = \sum_{n=-\infty}^{\infty} x[n]\,x[n-l], \qquad l = 0, \pm 1, \pm 2, \ldots \tag{1}$$

Comparing with the convolution sum shows that autocorrelation is the convolution of $x[n]$ with its folded copy (Experiment 1), and by the convolution theorem it is the inverse transform of the energy spectrum (the **Wiener-Khinchin** relation):

$$r_{xx}[l] = x[l] * x[-l] \quad \leftrightarrow \quad S_{xx}(e^{j\omega}) = |X(e^{j\omega})|^2 \tag{2}$$

Three properties follow directly and are verified numerically below:

$$r_{xx}[-l] = r_{xx}[l], \qquad |r_{xx}[l]| \le r_{xx}[0] = E_x = \sum_n \quad x^2[n] \tag{3}$$

For a finite record of N samples with mean $\overline{x}$, the normalised autocorrelation (autocorrelation function, ACF) is estimated by

$$\rho[l] = \frac{\hat{r}[l]}{\hat{r}[0]}, \qquad \hat{r}[l] = \frac{1}{N}\sum_{n=l}^{N-1}(x[n] - \overline{x})\,(x[n-l] - \overline{x}) \tag{4}$$

Dividing by N (biased) rather than $N - l$ (unbiased) gives a slightly smaller but far less noisy estimate at large lags, and is the standard choice. Two reference cases guide interpretation: a periodic signal of period P has an ACF that repeats with period P (a sinusoid gives $\rho[l] = \cos(2\pi l/P)$), while white noise has $\rho[l] \approx 0$ for $l \neq 0$, with 95 % of sample values inside

$$|\rho[l]| < \frac{1.96}{\sqrt{N}} \tag{5}$$

> **Note.** Autocorrelation is robust to noise because uncorrelated noise adds only to ρ[0]; the periodic part keeps reinforcing at its own lag. This is why a period can be read from the ACF even when the noise is larger than the signal (Section 7).

4. Dataset Description

The **international sunspot number** (version 2.0) is maintained by SILSO, the World Data Center for the sunspot index at the Royal Observatory of Belgium. It combines counts from about 80 observatories worldwide and has been recalibrated back to 1749, making it the longest continuous record of solar activity. The monthly mean series is a single 124 kB text file with one row per month, downloaded once and cached. The last 3 months are provisional. Sampling interval: T_s = 1 month, so 12 lags = 1 year.

Table 2. Dataset attributes

Property	Value	Property	Value
Provider	SILSO, Royal Observatory of Belgium	Series	monthly mean total sunspot number v2.0
Period	1749-01 to 2026-09	Samples N	3,333 months
Mean ± std	82.2 ± 67.6	Range	0 to 398.2
Largest month	1778-05	Months with observer count	2,504
Columns used	year, month, decimal year, SN	Licence	CC BY-NC 4.0 (SILSO)

Table 3. Most recent six rows of the dataset

Year	Month	Decimal year	SN	Std. dev.	Observations	Definitive
2026	04	2026.286	79.0	11.2	1345	yes
2026	05	2026.371	101.8	15.5	1319	yes
2026	06	2026.453	93.3	15.1	1316	yes
2026	07	2026.538	78.1	11.2	1371	provisional
2026	08	2026.623	76.0	12.3	1248	provisional
2026	09	2026.705	60.3	13.0	1117	provisional

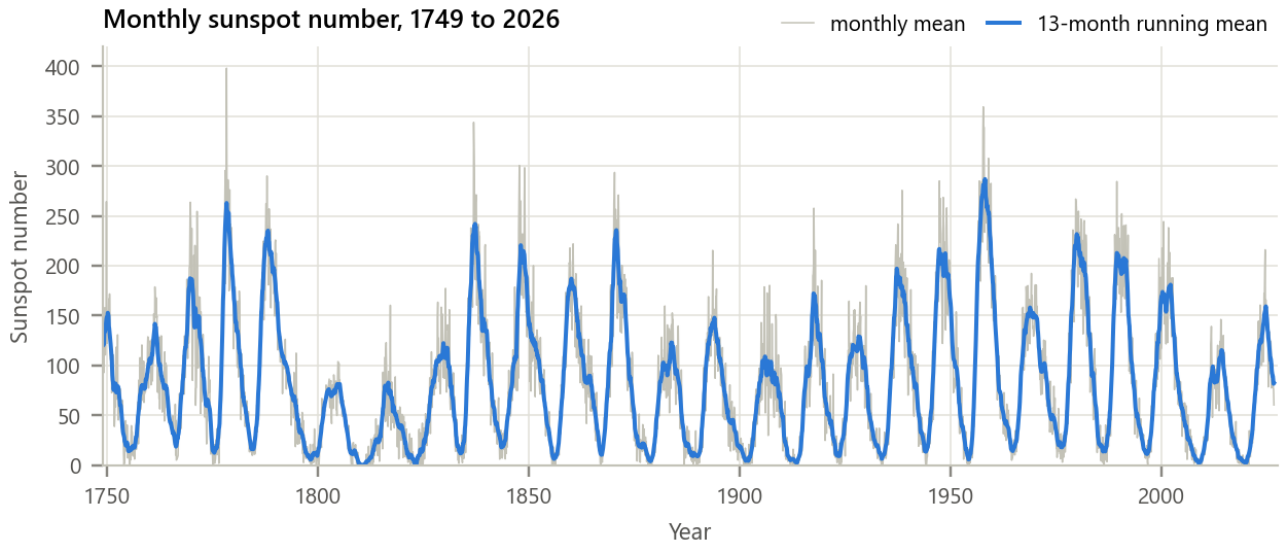


Figure 1. The monthly sunspot number and its 13-month running mean.

Inference. About 25 cycles are visible. Their heights vary widely (smoothed maxima of only about 80 in the early 1800s, the Dalton Minimum, but almost 300 in 1958), and their spacing is roughly, but not exactly, 11 years. The latest cycle (Cycle 25) peaked at a smoothed value of 159 around 2024.8. The monthly values scatter strongly around the smooth curve, so the period should be measured statistically rather than by eye.

5. Methodology and Implementation

The mean is removed, the biased ACF is computed up to 50 years of lag, and the period is taken as the largest ACF value between 60 and 200 months, refined by fitting a parabola through the peak and its neighbours. The same estimator is applied to four 70-year epochs, to the residual and the first difference, and to the record with added white noise.

Table 4. Software modules

Module	Responsibility
`config.py`	URL, lags, smoothing length, epochs, noise levels
`data_loader.py`	Download SILSO file once, cache, mark and fill missing months
`autocorr.py`	Three autocorrelation methods, biased / unbiased ACF, bounds, period finder
`analysis.py`	Main ACF, property checks, whiteness tests, epochs, noise robustness
`plots.py` / `gui.py`	Report figures / interactive periodicity detector
`test_autocorr.py`	10 unit tests (properties, sinusoid ACF = cosine, white-noise bound): all pass

Core of autocorr.py

```python
def autocorr_direct(x, max_lag):            # r[l] = sum x[n] x[n - l]
    return np.array([np.dot(x[l:], x[:len(x) - l]) for l in range(max_lag + 1)])

def autocorr_conv(x):                        # r[l] = x[l] * x[-l]
    return np.convolve(x, x[::-1])

def autocorr_fft(x, max_lag):                # Wiener-Khinchin: r = IDFT{|X|^2}
    nfft = 1 << int(np.ceil(np.log2(2 * len(x) - 1)))     # pad: no circular wrap
    return np.fft.irfft(np.abs(np.fft.rfft(x, nfft)) ** 2, nfft)[:max_lag + 1]

def acf(x, max_lag, unbiased=False):         # rho[l] = r[l] / r[0]
    x = x - x.mean()
    r = autocorr_fft(x, max_lag)
    r = r / (len(x) - np.arange(max_lag + 1)) if unbiased else r / len(x)
    return r / r[0]

```

GUI. `gui.py` (Figure 2) lets the user choose the raw, smoothed, residual or differenced series, any span of years, the maximum lag and the estimator, and add white noise of any strength. It shows the signal, its ACF with the 95 % band, the detected period with a significance check, and the first zero crossing. Run `python run_experiment.py` and `python gui.py`.

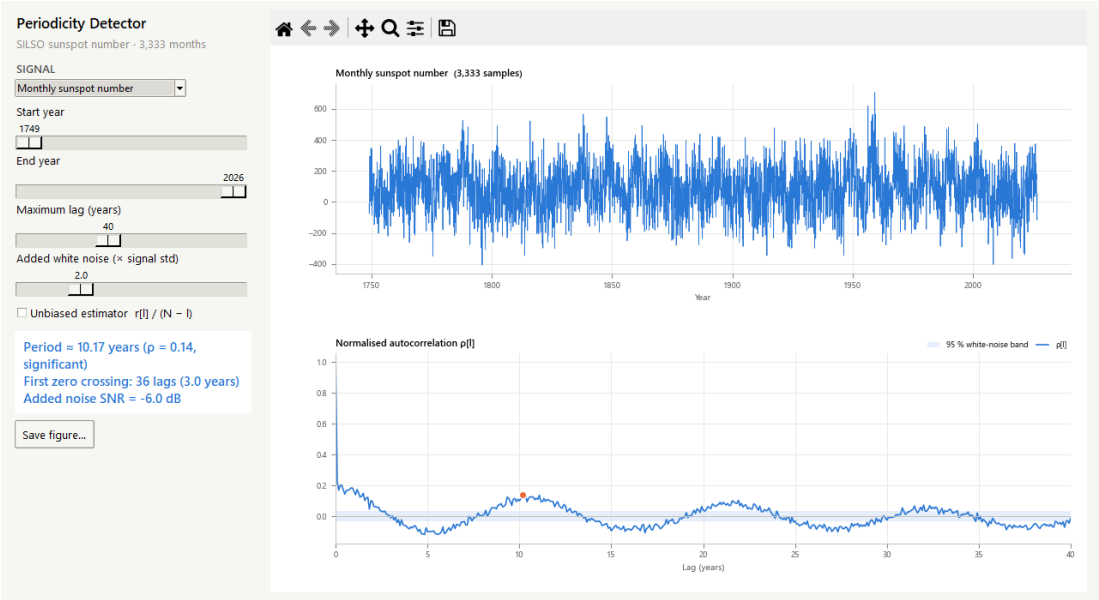


Figure 2. The periodicity detector with noise twice as strong as the signal (SNR −6 dB). The cycle is invisible in the time series but the ACF still peaks at about 10 years, above the band.

6. Calculations

Hand example. For x[n] = {2, −1, 3, 1} (n = 0 … 3), each lag multiplies the sequence with a shifted copy and adds the overlapping products:

Table 5. Autocorrelation by hand

Lag l	Overlapping products x[n]·x[n − l]	r[l]	ρ[l] = r[l]/r[0]
0	2·2 + (−1)(−1) + 3·3 + 1·1	15	1.000
1	(−1)·2 + 3·(−1) + 1·3	-2	-0.133
2	3·2 + 1·(−1)	5	0.333
3	1·2	2	0.133

By symmetry r[−1] = -2, r[−2] = 5, r[−3] = 2, so the full sequence has 2N − 1 = 7 values with its maximum r[0] = 15 = E_x at the centre. **For the sunspot record:** the 95 % white-noise limit is 1.96/√3333 = 0.0339; the ACF peak at lag 128 with parabolic refinement gives 127.8 months ÷ 12 = **10.65 years**. Adding noise with standard deviation 3σ gives SNR = 10 $\log_{10}(1/3^2)$ = -9.5 dB. The program reproduces the hand values exactly (checked by `test_autocorr.py`).

7. Results and Discussion

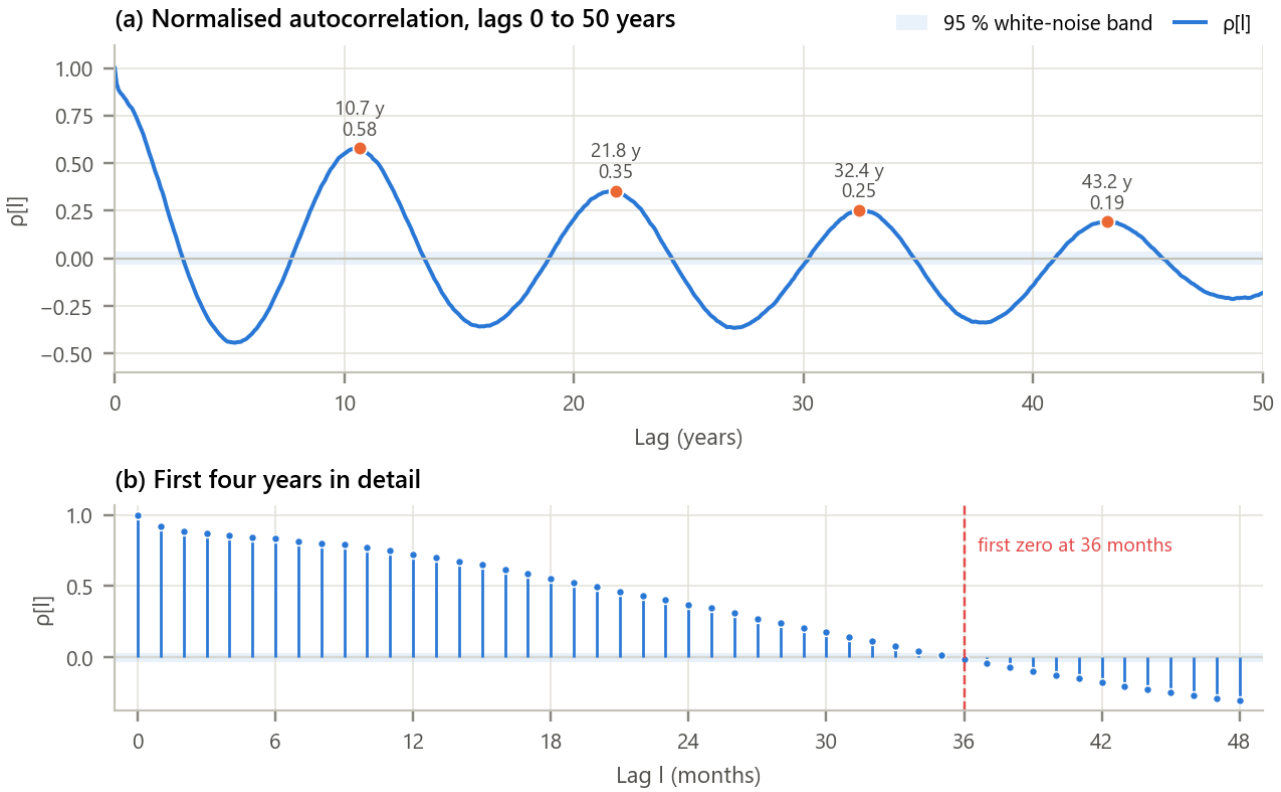


Figure 3. Normalised autocorrelation of the monthly sunspot number. (a) Lags up to 50 years with the 95 % white-noise band; (b) the first four years.

Inference.

- The ACF oscillates like a damped cosine. Its first peak is at **10.65 years** ($\rho = 0.579$), far above the ± 0.034 band, so the 11-year cycle is overwhelmingly significant. Further peaks at 21.8, 32.4, 43.2 years are integer multiples of the period.
- The peaks shrink (0.58, 0.35, 0.25, 0.19) because cycles differ in length and height, so the signal is only quasi-periodic. The deepest trough ($\rho = -0.44$ at 5.2 years) is half a cycle: maximum is followed by minimum.
- Neighbouring months are strongly related ($\rho[1] = 0.918$, $\rho[12] = 0.724$); the correlation falls below 0.5 after 20 months and crosses zero at 36 months, about a quarter of the cycle. Persistence forecasts are therefore useful for one to two years only.

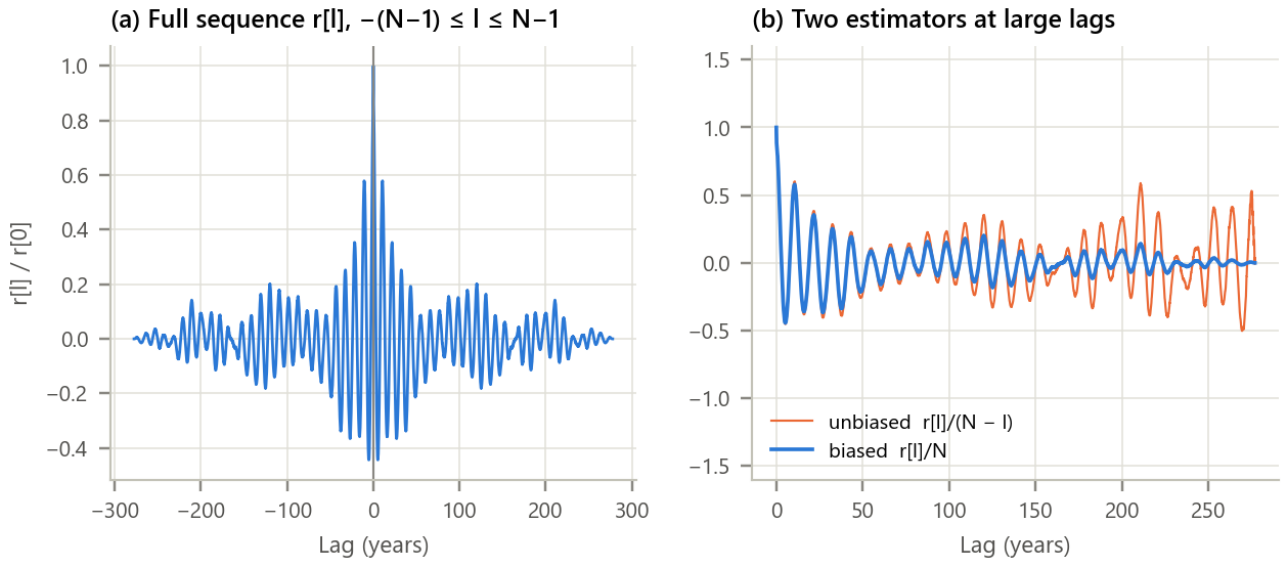


Figure 4. Properties of the autocorrelation sequence. (a) The full sequence for negative and positive lags; (b) biased versus unbiased estimates.

Inference. The full sequence is exactly even (maximum asymmetry 0e+00) and never exceeds its value at l = 0, which equals the signal energy (difference 0e+00). The direct, convolution and FFT methods agreed to 4e-16. At large lags the unbiased estimate swings wildly, because it averages only the few remaining overlapping samples, while the biased estimate tapers smoothly to zero; this is why the biased estimator is used.

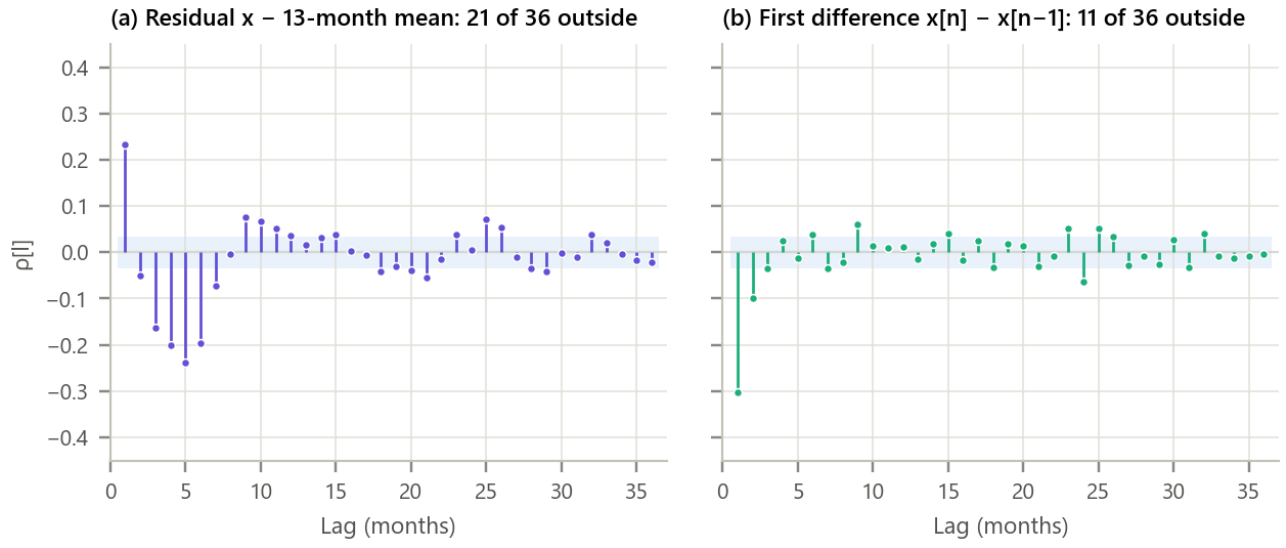


Figure 5. Testing the leftover variation for whiteness. ACF of (a) the residual after removing the 13-month mean and (b) the month-to-month difference.

Inference. Neither is white noise. The residual has 21 of 36 lags outside the 95 % band (where about 2 would be expected by chance), with a positive ρ[1] = 0.23 and a negative dip at 3 to 6 months, partly created by the 13-month smoothing itself. The difference is close to white except for ρ[1] = -0.30: a high month tends to be followed by a lower one, the signature of independent counting errors added to a slowly varying signal. The ACF therefore tells us which kind of model the noise needs.

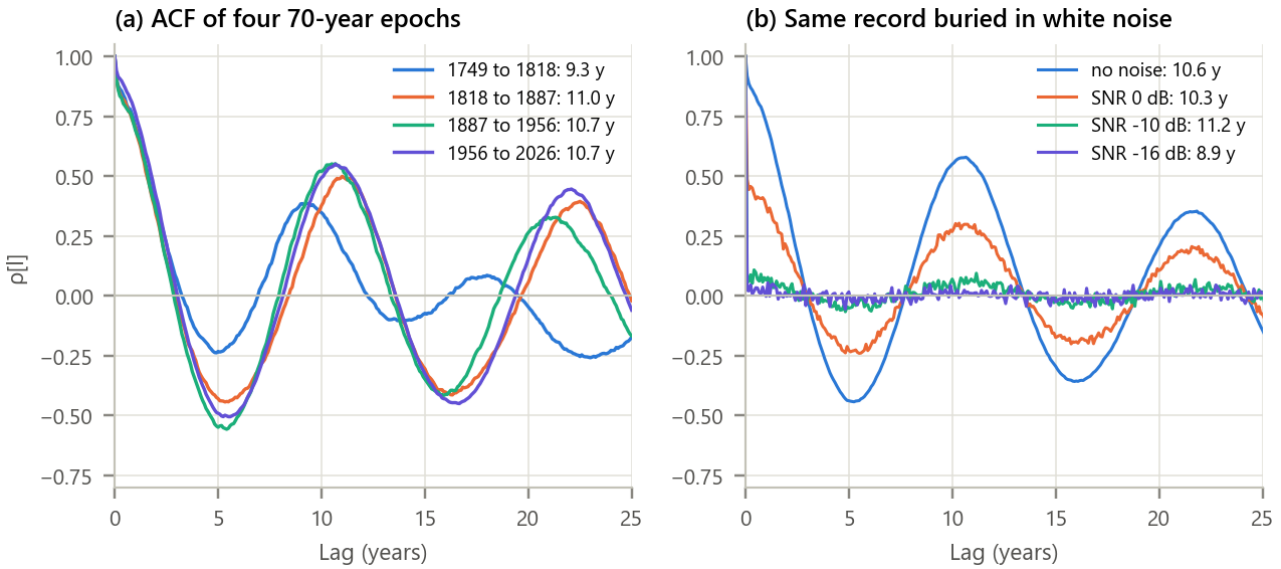


Figure 6. Stability and robustness of the period estimate. (a) ACF of four 70-year epochs; (b) the full record with added white noise.

Inference.

- Since 1818 the period is stable at 11.0, 10.7 and 10.7 years, but 1749 to 1818 gives only 9.3 years with a weak peak ($\rho = 0.39$): that epoch contains short cycles and the Dalton Minimum, and its early counts are the least reliable.
- With noise as strong as the signal (0 dB) the period is still 10.3 years; at -10 dB it is 11.2 years with a small but significant peak (0.094). At -16 dB the peak (0.046) sinks to the noise band (0.034) and the estimate (8.9 years) is unreliable.

Table 6. Summary of results

Quantity	Result
Solar-cycle period from the ACF	10.65 years ($\rho = 0.579$)
Multiples of the period	21.8 y, 32.4 y, 43.2 y
Decorrelation: $\rho < 0.5$ / first zero	20 / 36 months
Period by epoch	1749 to 1818: 9.3 y, 1818 to 1887: 11.0 y, 1887 to 1956: 10.7 y, 1956 to 2026: 10.7 y
Whiteness: residual / difference outside band	21 / 11 of 36 lags
Lowest SNR with a reliable period	-10 dB
Next maximum, rough (last max + period)	about 2035

8. Conclusion

Autocorrelation extracted the solar cycle directly from the data: a period of 10.65 years, with decaying repeats that quantify how irregular the Sun is. The theoretical properties (even symmetry, maximum at zero lag equal to the energy, equivalence of the direct, convolution and spectral methods) were confirmed to machine precision. The ACF also served as a diagnostic: the cycle was stable at about 10.7 to 11 years for two centuries, the leftover month-to-month variation is not white noise, and periodicity was still recoverable at an SNR of -10 dB. Combined with the most recent maximum (2024.8), the period suggests the next

maximum around 2035, a rough but useful planning horizon for satellite and grid operators; real forecasts refine it with physical models.

9. Learning Outcomes

- Compute autocorrelation by definition, by convolution with the folded signal and via the power spectrum.
- Verify and use its properties: even symmetry, maximum at zero lag, r[0] = energy.
- Detect and measure hidden periodicity, and choose between biased and unbiased estimators.
- Test a sequence for whiteness with the $\pm 1.96/\sqrt{N}$ band and judge the noise robustness of a period estimate.

10. References

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[2] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Pearson, 2010, ch. 10.

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[5] G. E. P. Box, G. M. Jenkins and G. C. Reinsel, *Time Series Analysis*, 4th ed. Wiley, 2008, ch. 2.